

I/O

(reference: text Sections 5.2, 5.11)

wide diversity of I/O devices

- storage devices such as magnetic disks, SSDs, USB flash drives, ...
- “input” and/or “output” devices that facilitate communication with users or other machines

performing I/O

service provided by **operating system** (operating system provides *protected sharing*, handles low-level details)

three basic approaches:

- **polling**: processor initiates or enables I/O operation, then periodically asks I/O device "are you done yet?"
- **interrupt-driven I/O**: processor initiates or enables I/O, I/O device **interrupts** processor when I/O operation completed
- **DMA** (“**direct memory access**”): like interrupt-driven I/O, except that **DMA controller** is able to **directly transfer data to/from memory**